Project Proposal

Petras Paul Friedel Luca Baumgartner Luis Reck Colin

2026-06-08

Table of contents

1 Dataset 1

2 Research Questions 1

3 Initial EDA 2

4 Analysis Plan 6

5 Workflow organisation 7
5.1 Code style 7
5.2 Packages 7
5.3 Git workflow 7
5.4 .gitignore 8
5.5 Weights 8

1 Dataset

Source: https://search.gesis.org/research_data/ZA5280 License: Open Source

2 Research Questions

1. To what extent do economic fluctuations (measured by GDP, growth, or a decline in GDP) influence voter turnout and the election results of the governing parties in German federal elections? Pushed back: (2. To what extent do the objective economic situation (net household income) and the subjective expectation regarding one's own

future financial situation influence support for state income redistribution in Germany during the pandemic year of 2021?)

3 Initial EDA

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.2
v ggplot2   4.0.0     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.1.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```
library(here)
```

```
Warning: Paket 'here' wurde unter R Version 4.5.3 erstellt
```

```
here() starts at C:/Users/PaulP/OneDrive/Dokumente/PP/Studium/Bachelor Informatik/4. Semeste
```

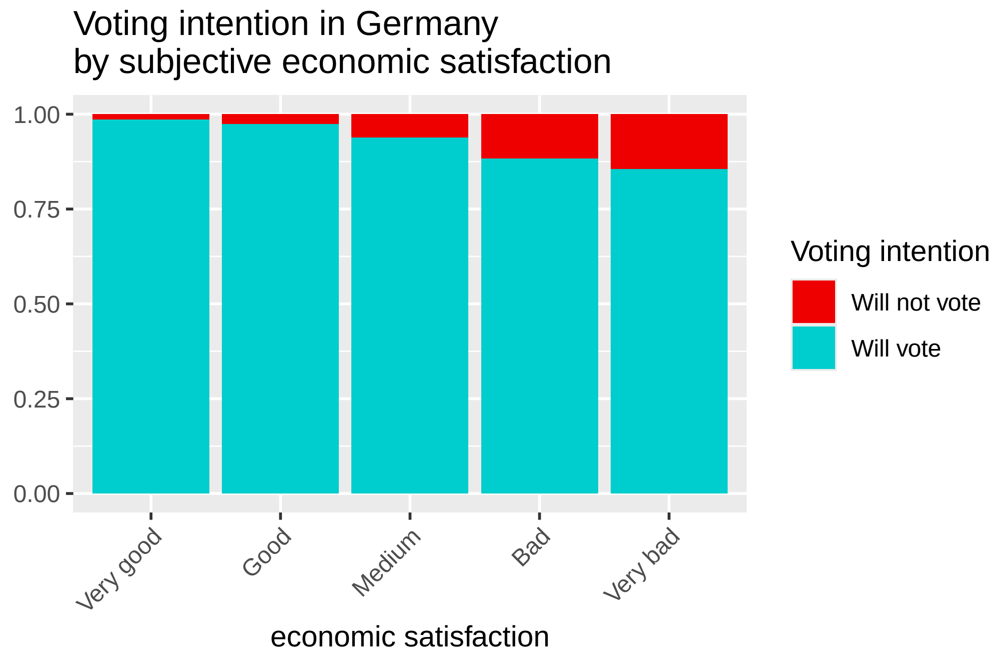
```
library(haven)
allbus_mini_clean <- read_dta(here("data", "processed", "allbus_mini_clean.dta"))
```

To continue, please run 01, 02, and 03 in the code folder.

```
# First plot.

# Initial sorting of the column names, so we can find the proper variables easier.
alph_allbus_mini_clean <- allbus_mini_clean[, order(colnames(allbus_mini_clean))]
alph_allbus_mini_clean %>%
  # Filtering only proper values for the ep01 variable. (personal economic evaluation)
  filter(ep01 %in% c(1, 2, 3, 4, 5)) %>%
  # Filtering only the proper values for the pv01 variable. (voting intention)
  # we remove don't know, no answer, answer rejected and not eligible to vote.
```

```r
filter(pv01 %in% c(1, 2, 3, 4, 6, 42, 90, 91)) %>%
# we create our own voting variable and make it able to compute expected turnout.
mutate(
  voting_intention = if_else(pv01 %in% c(1, 2, 3, 4, 6, 42, 90), 1, 0),
  voting_intention = as.factor(voting_intention)
) %>%
# ep01 as factor
mutate(ep01 = as.factor(ep01)) |>
# recoded ep01
mutate(ep01 = recode(ep01,
  "1" = "Very good",
  "2" = "Good",
  "3" = "Medium",
  "4" = "Bad",
  "5" = "Very bad"
)) |>
# we remove all unnecessary variables, but keeping the weight is very important.
select(ep01, pv01, voting_intention, wghtpew) %>%
# We can now plot the weighted result.
ggplot(aes(x = ep01, fill = voting_intention, weight = wghtpew)) +
geom_bar(position = "fill") +
# labels.
labs(
  title = "Voting intention in Germany \nby subjective economic satisfaction",
  x = "economic satisfaction",
  y = "",
  fill = "Voting intention"
) +
scale_fill_manual(values = c("red2", "cyan3"), labels = c("Will not vote", "Will vote")) +
theme(axis.text.x = element_text(
  angle = 45,
  vjust = 1,
  hjust = 1
))
```



As we can see, people with are more negative opinion on the economic situation in Germany are more likely to announce that they would not be voting.

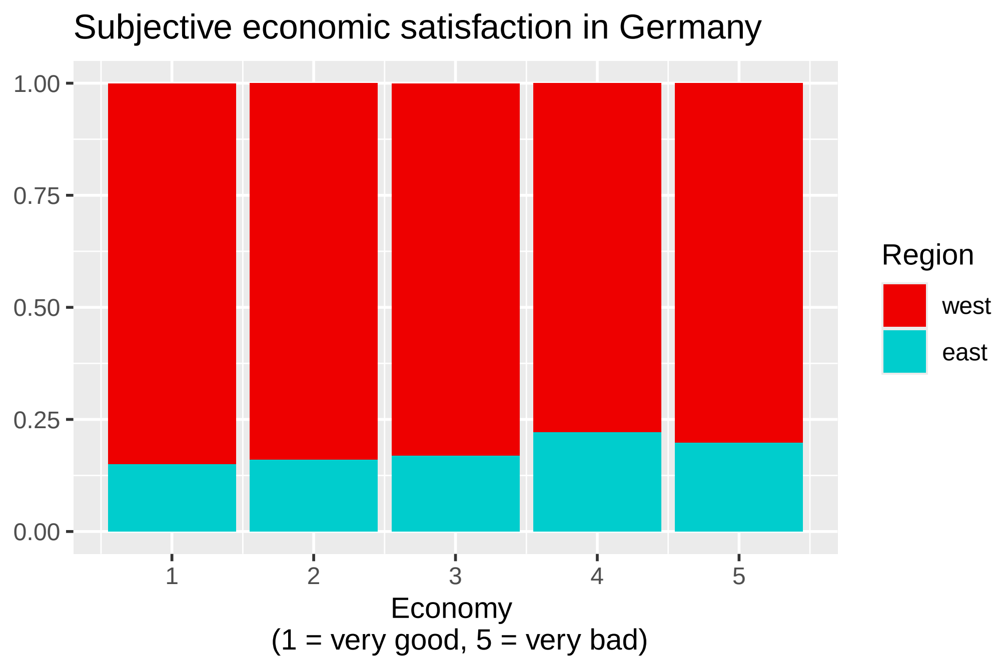
```
alph_allbus_mini_clean %>%
  group_by(pv01) %>%
  summarise("Total amount" = n()) |>
  rename("Voting intention" = pv01)
```

```
# A tibble: 13 x 2
   `Voting intention`          `Total amount`
   <dbl+lbl>                            <int>
 1 -50 [NICHT WAHLBERECHTIGT]             433
 2 -42 [DATENFEHLER: MFN]                  79
 3  -9 [KEINE ANGABE]                     413
 4  -8 [WEISS NICHT]                     1558
 5  -7 [VERWEIGERT]                       750
 6   1 [CDU-CSU]                         2823
 7   2 [SPD]                             1929
 8   3 [FDP]                             1080
 9   4 [DIE GRUENEN]                     2182
10   6 [DIE LINKE]                        839
11  42 [AFD]                             1003
12  90 [ANDERE PARTEI]                    391
```

```
13  91 [WUERDE NICHT WAEHLEN]          585
```

```
# Very quick overview on how people answered the voting intention question.
```

```
alph_allbus_mini_clean %>%
  filter(ep01 %in% c(1, 2, 3, 4, 5)) %>%
  mutate(eastwest = as.factor(eastwest)) %>%
  ggplot(aes(x = ep01, fill = eastwest, weight = wghtpew)) +
  geom_bar(position = "fill") +
  labs(
    x = "Economy \n (1 = very good, 5 = very bad)",
    y = "",
    fill = "Region",
    title = "Subjective economic satisfaction in Germany"
  ) +
  scale_fill_manual(labels = c("west", "east"), values = c("red2", "cyan3"))
```



As we can see here, East Germans are more likely to have a negative view on Germany's economic situation than West Germans.

4 Analysis Plan

1. Research question

- Data basis: ALLBUS 1980-2023, linking macroeconomic time series (Destatis GDP growth) covering the election year.
- Descriptive: Frequency distributions of the DVs, means/SDs of the IV across election years, cross-tabulations of region × voting behavior.
- Bivariate: Correlations between GDP growth and aggregated voter turnout/governing-party vote and the change compared to the previous election cycle; chi^2 tests for categorical control variables.
- Multivariate:
  - Logistic regression 1: Voter turnout ~ GDP growth + control variables
  - Logistic regression 2: Vote for governing party ~ GDP growth + control variables

- Robustness: Interaction effects (GDP × region; GDP × employment status) to test whether economic voting is context-dependent. – Interpretation: Effect sizes (odds ratios, average marginal effects) rather than mere significance tests.

2. Question

- Data basis: ALLBUS 2021
- Descriptive: Distribution of the DV, income deciles, distribution of future expectations; separately for East/West.
- Bivariate:
  - Correlation income support for redistribution (expected: negative)
  - Correlation expectation support for redistribution (expected: more pessimistic expectation → higher support)
  - Multivariate — hierarchical linear regression:
  - Model 1: DV ~ control variables
  - Model 2: + IV1 (objective income)
  - Model 3: + IV2 (subjective expectation)
  - Comparison of ΔR^2 → which block explains more variance

- Interaction: Income × expectation, to test whether, for instance, high-income but pessimistic individuals become particularly supportive of redistribution ("fear of falling" hypothesis).
- Robustness: Sensitivity analysis with log-transformed income; separate estimations for East/West.

5 Workflow organisation

5.1 Code style

We will follow the Tidyverse Style Guide. Code formatting will be handled using the styler package, ideally configured to format code automatically on save. In addition, we will enforce consistent formatting through a GitHub Actions workflow that checks whether the code complies with the defined style guide.

5.2 Packages

As of now we will use following packages in our project:

- tidyverse
- here
- haven
- janitor
- glue
- readr
- lubridate
- withr

If we don't actively encounter version incompatibility problems, we will probably not use package version locks as we're assuming that package versions are generally backwards compatible.

5.3 Git workflow

The main idea behind branching is to enable a group of people to work simultaneously on the same project without interfering with each other, while also partitioning features or logically related steps to improve clarity and flexibility. This is exactly how we will use it.

We follow a feature-based branching strategy, where each step or logically related set of steps has its own branch. These branches are accessible to all team members if needed; however, to prevent conflicts, we generally restrict work on a given branch to one team member at a time.

5.4 .gitignore

What we will exclude from git is user specific data, because it is irrelevant for others and potentially private, as well as build artefacts and OS noise, as it is irrelevant to others as well.

What we will not exclude is all relevant working data, informational files, as well as environmental files.

5.5 Weights

If we need to make graphics that are supposed to be represent Germany accurately, we will use the adequate weighting variable from the data set (I.e. wghtpew from allbus). Other data sets like the polling from dawum already use their own weights and provide the weighted means in their polling numbers.